

4.9.4 The Transmission Control Protocol/Internet Protocol (TCP/IP) protocol

4.9.4.1 TCP/IP

Content	Additional information
Describe the role of the four layers of the TCP/IP stack (application, transport, network, link).	
Describe the role of sockets in the TCP/IP stack.	
Be familiar with the role of MAC (Media Access Control) addresses.	
Explain what the well-known ports and client ports are used for and the differences between them.	

4.9.4.2 Standard application layer protocols

Content	Additional information
Be familiar with the following protocols: • FTP (File Transfer Protocol) • HTTP (Hypertext Transfer Protocol) • HTTPS (Hypertext Transfer Protocol Secure) • POP3 (Post Office Protocol (v3)) • SMTP (Simple Mail Transfer Protocol) • SSH (Secure Shell).	
Be familiar with FTP client software and an FTP server, with regard to transferring files using anonymous and non-anonymous access.	
Be familiar with how SSH is used for remote management.	

Content	Additional information
Know how an SSH client is used to make a TCP connection to a remote port for the purpose of sending commands to this port using application level protocols such as GET for HTTP, SMTP commands for sending email and POP3 for retrieving email.	
Be familiar with using SSH to log in securely to a remote computer and execute commands.	
Explain the role of an email server in retrieving and sending email.	
Explain the role of a web server in serving up web pages in text form.	
Understand the role of a web browser in retrieving web pages and web page resources and rendering these accordingly.	

4.9.4.3 IP address structure

Content	Additional information
Know that an IP address is split into a network identifier part and a host identifier part.	

4.9.4.4 Subnet masking

Content	Additional information
Know that networks can be divided into subnets and know how a subnet mask is used to identify the network identifier part of the IP address.	

4.9.4.5 IP standards

Content	Additional information
Know that there are currently two standards of IP address, v4 and v6.	
Know why v6 was introduced.	

4.9.4.6 Public and private IP addresses

Content	Additional information
Distinguish between routable and non-routable IP addresses.	

4.9.4.7 Dynamic Host Configuration Protocol (DHCP)

Content	Additional information
Understand the purpose and function of the DHCP system.	

4.9.4.8 Network Address Translation (NAT)

Content	Additional information
Explain the basic concept of NAT and why it is used.	

4.9.4.9 Port forwarding

Content	Additional information
Explain the basic concept of port forwarding and why it is used.	

4.9.4.10 Client server model

Content	Additional information
Be familiar with the client server model.	Client sends a request message to server, server responds to request by replying with a response message to client.
Be familiar with the Websocket protocol and know why it is used and where it is used.	The Websocket specification defines an API (Application Programming Interface) establishing a full-duplex 'socket' connection between a web browser and a server over TCP. This means that there is a persistent connection between client and server, allowing both parties to send data at any time.

Content	Additional information
Be familiar with the principles of Web CRUD Applications and REST: • CRUD is an acronym for: • C – Create • R – Retrieve • U – Update • D – Delete. • REST enables CRUD to be mapped to database functions (SQL) as follows: • GET → SELECT • POST → INSERT • DELETE → DELETE • PUT → UPDATE.	Students should understand the principles: • database connected to browser using REST – Representational State Transfer - which relies on HTTP request methods • REST allows JavaScript to talk to server through HTTP • REST API (Application Programming Interface) created and run on server, browser Javascript calls API • JSON (JavaScript Object Notation) or XML can be used to transmit data between a server and web application • Javascript referenced by HTML file, eg index.html, is run in browser.
Compare JSON (Java script object notation) with XML.	JSON compared with XML is: • easier for a human to read • more compact • easier to create • easier for computers to parse and therefore quicker to parse.

4.9.4.11 Thin- versus thick-client computing

Content	Additional information
Compare and contrast thin-client computing with thick-client computing.	